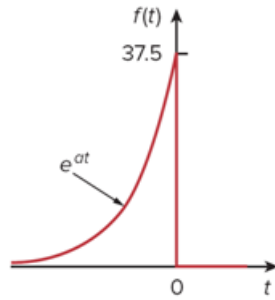


## Problem

Determine the Fourier transform of the “switched-off” exponential function in Fig. 18.8.

**Figure 18.8**



## Step-by-step solution

## Step 1 of 2

Refer to the exponential function in Figure 18.8 in the textbook.

The exponential function is expressed as,

$$f(t) = \begin{cases} 37.5e^{at}, & t \leq 0 \\ 0, & t > 0 \end{cases}$$

Find the Fourier transform of the “switched-off” exponential function.

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \\ &= \int_{-\infty}^0 37.5e^{at} e^{-j\omega t} dt \\ &= \int_{-\infty}^0 37.5e^{(a-j\omega)t} dt \end{aligned}$$

## Step 2 of 2

Simplify the integration.

$$\begin{aligned} F(\omega) &= 37.5 \left[ \frac{e^{(a-j\omega)t}}{(a-j\omega)} \right]_{-\infty}^0 \\ &= \frac{37.5}{a-j\omega} [e^0 - e^{-\infty}] \\ &= \frac{37.5}{a-j\omega} [1-0] \\ &= \frac{37.5}{a-j\omega} \end{aligned}$$

Thus, the Fourier transform of the “switched-off” exponential function is  $\boxed{\frac{37.5}{a-j\omega}}$ .